Comparative Analysis of Machine Learning Classification Algorithms with PCA for Network Intrusion Detection

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Abstract

Network intrusion detection aims to distinguish legitimate network activity from malicious traffic in high-dimensional flow environments. This study evaluates five classification architectures—Multilayer Perceptron (MLP), Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), Logistic Regression (LR), and a continuous baseline of Rounded Linear Regression—paired with Principal Component Analysis (PCA) on selected records from the CIC-IDS2017 benchmark dataset. The workflow aggregates multi-day packet captures, encodes categorical features, mean-imputes missing/infinite values, and applies PCA at 5, 10, and 15 components across three train-test partitioning ratios (80:20, 60:40, and 40:60), establishing a systematic matrix of 45 experimental configurations. In the primary 80:20 evaluation suite spearheaded by **Shashwat Bikram Singh**, the 15-component MLP achieved the study's highest overall accuracy at **99.42%** (with a weighted F1 score of **99.38%**), outperforming the corresponding 60:40 configuration (99.19% accuracy, 99.12% F1). The empirical results demonstrate that maximizing training volume significantly stabilizes orthogonal PCA subspace estimation and non-linear boundary construction for minority attack categories (e.g., Web Attacks), while models such as QDA and Logistic Regression suffer from severe precision-recall disparity on imbalanced class distributions.

Keywords: *CIC-IDS2017 • Principal Component Analysis (PCA) • Network Intrusion Detection • Multilayer Perceptron • Machine Learning Benchmark*

1. Introduction & Methodology

Modern Network Intrusion Detection Systems (NIDS) must reliably separate legitimate network packets from sophisticated cyber threats in real time. Because raw flow records contain up to 78 statistical dimensions, dimensionality reduction via Principal Component Analysis (PCA) is commonly used to accelerate inference. To analyze performance boundaries across feature dimensions and data partitions, a 45-configuration benchmark matrix was conducted across three data partitions: **80:20 Split Suite (SDGC_60_40_0.2)** led by Shashwat Bikram Singh, **60:40 Split Suite (SDGC_60_40_0.4)** led by Ashutosh Kumar Jha, and **40:60 Split Suite (SDGC_40_60_0.6)** led by Siddhant Dixit.

2. Team Division of Work & Experimental Regimes

- **Shashwat Bikram Singh (25BTRCN173):** Designed and executed the **80:20 train-test split experimental suite (SDGC_60_40_0.2)** across 15 model configurations (PCA 5, 10, 15 components × 5 algorithms). Led NaN/Inf imputation logic, categorical label encoding, confusion matrix metrics, and cross-split comparative evaluation.
- **Ashutosh Kumar Jha (25BTRCN139):** Implemented and evaluated the **60:40 train-test split suite (SDGC_60_40_0.4)**, evaluating intermediate PCA dimensionalities and precision-recall trade-offs.
- **Siddhant Dixit (25BTRCN112):** Project coordination, repository infrastructure, and implementation of the **40:60 train-test split suite (SDGC_40_60_0.6)**.

3. Consolidated Experimental Benchmark Results

Model	Split	PCA	Accuracy	Precision	Recall	F1 Score	Lead Contributor
MLP	80:20	15	**99.42%**	99.41%	99.42%	**99.38%**	**Shashwat B. Singh**
MLP	80:20	10	**99.28%**	99.27%	99.28%	99.24%	**Shashwat B. Singh**
MLP	60:40	15	99.19%	99.19%	99.19%	99.12%	A. Jha / S. Dixit

MLP	60:40	10	99.05%	99.06%	99.05%	98.99%	A. Jha / S. Dixit
MLP	80:20	5	**99.01%**	98.98%	99.01%	98.95%	**Shashwat B. Singh**
MLP	60:40	5	98.77%	98.78%	98.77%	98.70%	A. Jha / S. Dixit
LDA	80:20	15	**86.82%**	89.95%	86.82%	87.80%	**Shashwat B. Singh**
LDA	60:40	15	85.65%	89.09%	85.65%	86.75%	A. Jha / S. Dixit
LDA	80:20	10	85.34%	89.10%	85.34%	86.42%	**Shashwat B. Singh**
Logistic Reg.	80:20	15	81.35%	65.80%	81.35%	72.40%	**Shashwat B. Singh**
QDA	80:20	10	68.15%	91.20%	68.15%	76.80%	**Shashwat B. Singh**
QDA	60:40	10	66.43%	90.67%	66.43%	75.03%	A. Jha / S. Dixit

4. Top Study Configurations & Findings

1. Rank 1: MLP (80:20, PCA 15) — 99.42% Accuracy (Lead: Shashwat Bikram Singh)
2. Rank 2: MLP (80:20, PCA 10) — 99.28% Accuracy (Lead: Shashwat Bikram Singh)
3. Rank 3: MLP (60:40, PCA 15) — 99.19% Accuracy (Lead: Ashutosh Kumar Jha / Siddhant Dixit)
4. Rank 4: MLP (60:40, PCA 10) — 99.05% Accuracy (Lead: Ashutosh Kumar Jha / Siddhant Dixit)
5. Rank 5: MLP (80:20, PCA 5) — 99.01% Accuracy (Lead: Shashwat Bikram Singh)

The 80:20 Split Advantage: The 80:20 split suite developed by Shashwat Bikram Singh achieved the top 2 overall study scores. The additional 20% training partition significantly bolstered representation of long-tail attack classes (e.g. Web Attacks), preventing hyperplane degradation and improving overall multi-class recall. MLP demonstrated remarkable resilience, retaining 99.01% accuracy even under a 93.6% dimensionality compression (5 components).

5. Author Contributions Statement

Shashwat Bikram Singh (25BTRCN173): Conceptualized high-training split regime; implemented and evaluated the entire 80:20 split suite (SDGC_60_40_0.2); designed data imputation and encoding; authored 80:20 performance analysis and comparative discussion.

Ashutosh Kumar Jha (25BTRCN139): Implemented and evaluated the 60:40 split suite (SDGC_60_40_0.4); conducted intermediate PCA and precision-recall trade-off evaluations.

Siddhant Dixit (25BTRCN112): Project coordination; repository infrastructure; implemented the 40:60 split suite (SDGC_40_60_0.6); performed initial dataset acquisition.